M. SYED MARJUK

EIA Specialist | Marine & Terrestrial Biologist | Remote Sensing & AI - ML Modelling

Project Fellow – Indian Space Research Organisation (ISRO)

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PROFESSIONAL PROFILE

Environmental Impact Assessment (EIA) Specialist and interdisciplinary Marine Biologist with **6+ years of hands-on experience** in ecological baseline surveys, marine biodiversity assessments, and regulatory EIA reporting as per Indian MoEF&CC norms. Has served as **EIA Biologist on three major projects — Ennore Oil Spill Impact Assessment (CPCL), Karaikal Port Baseline Survey, and TNEB Tuticorin & Tirunelveli Baseline Survey** (Chola MS Risk Services). Expert taxonomist from Protozoa to Mammalia, with specialist-level identification of marine plankton, avifauna, reptiles, mammals, and marine invertebrates. Trained ornithologist (SACON, MoEF&CC) with field experience in mist netting, bird banding, radio-tagging, and camera traps. ISRO-SAC-funded researcher with 7 peer-reviewed publications (93 citations, h-index: 3), and proficiency in GIS/remote sensing and machine learning tools for environmental monitoring.

RESEARCH METRICS

Google Scholar	7 Peer-Reviewed Articles | 93 Citations | h-index: 3 | i10-index: 2
ORCID	0000-0002-0834-3177
Scopus / WoS	Publications indexed in Elsevier (*Aquaculture, Regional Studies in Marine Science, Chemosphere, Marine Pollution Bulletin*), Springer (*Biomass conversion and Biorefinery*)

OCEANOGRAPHIC RESEARCH CRUISE EXPERIENCE

7 Research Cruises | Chief Scientist on 2 Research Cruises | Bay of Bengal & Arabian Sea

Vessels	ORV Sagar Manjusha | RV Sindhu Sankalp | CRV Sagar Tara | CRV Sagar Anveshika
Programme	ISRO OceanSAT-3 / SAC Satellite Validation — OCM-3 Ocean Colour Products

- ❖ Participated in 7 oceanographic research cruises as part of the ISRO OceanSAT-3 satellite validation programme; acted as Chief Scientist on 2 independent cruises, overseeing scientific operations, station planning, data integrity, and team coordination.
- ❖ Executed systematic plankton net sampling (WP-2 vertical hauls, oblique tows) across multiple oceanic stations in the Bay of Bengal and Arabian Sea, preserved and enumerated phytoplankton and zooplankton samples for biomass estimation and taxonomic analysis.
- ❖ Collected in-situ oceanographic parameters at each sampling station: sea surface temperature (SST), salinity, dissolved oxygen, nutrients (nitrate, phosphate, silicate), and chlorophyll-a; integrated field data with OCM-3 and MODIS-Aqua satellite imagery for algorithm validation.
- ❖ Deployed CTD profilers, Niskin bottles, and Grap samplers; maintained full chain-of-custody documentation and field data logging protocols in direct coordination with ISRO-SAC scientists.
- ❖ Operated GPS/GNSS instruments for real-time georeferenced station positioning and GIS data layer generation across all cruises.

ACADEMIC CHRONICLE

PhD Thesis:

"Application of Advanced Remote Sensing for Plankton Biomass Monitoring in the Southwestern Bay of Bengal and AI-Enhanced Predictive Modelling for the Sustainable Copepod Production"

ISRO-SAC Funded | Bharathidasan University, Tiruchirappalli | 2023–2026 | Thesis Submitted

Qualification	College	Board / University	Year	Aggregate (%)
Ph.D. Zoology (Inter-Disciplinary **Marine Science**)	Bharathidasan University, Tiruchirappalli	Bharathidasan University	2026	Thesis Submitted
M.Sc. Zoology	Hajee Karutha Rowther Howdia College	Madurai Kamaraj University	2021	First Class with Distinction
B.Sc. Zoology	Hajee Karutha Rowther Howdia College	Madurai Kamaraj University	2018	First Class
HDCA (Honours Diploma in Computer Application)	Computer Software College	Private	2019	First Class with Outstanding

PROFESSIONAL EXPERIENCE

Senior Research Fellow — Centre for Oceanic AI

Chinmaya Vishwa Vidyapeeth (CVV)

July 2026 – Present | Onakkoor, Kerala, India

- Contribute to interdisciplinary research integrating marine science, AI/ML, image analytics, and environmental data science for ocean observation and monitoring.
- Support the development, validation, and field deployment of AI-based marine imaging systems, including automated zooplankton detection and identification.
- Conduct marine field sampling, biological data collection, image/data analysis, and validation of AI-assisted oceanographic technologies.
- Publish research in peer-reviewed Scopus-indexed journals, guide student research projects, and contribute to research proposals, collaborations, and scientific project development.

Project Associate — Marine Science & Remote Sensing (ISRO OceanSAT-3)

Marine Planktonology & Aquaculture Laboratory, Dept. of Marine Science, Bharathidasan University

March 2023 – March 2026 | Tiruchirappalli, India

Funded by ISRO, Space Applications Centre (SAC)

- Developed machine learning and ANN models (MATLAB Regression Learner — SVM, GPR, Ensemble Trees, ANN) to predict phytoplankton biomass (PBM) and zooplankton biomass (ZBM) from satellite-derived SST and chlorophyll-a; validated predictions against in-situ plankton field data.
- Translated trained ANN architectures into mathematical band-math expressions for pixel-wise deployment in SeaDAS, enabling scalable satellite-based biomass mapping across the Bay of Bengal.

- ❖ Collaborated directly with ISRO-SAC scientists on OCM-3 ocean colour remote sensing product development and algorithm validation.
- ❖ Managed integration of in-situ water quality, nutrient, pigment, and plankton datasets with MODIS-Aqua and OCM-3 imagery in SeaDAS and QGIS.

Environmental Impact Assessment Biologist — Ennore Oil Spill Response

Chennai Petroleum Corporation Limited (CPCL) & TERI, Delhi — Field Consultancy

June 2022 – February 2023 | Ennore (Chennai) & Karaikal, India

- ❖ Acted as Marine Biologist to Ennore and Karaikal oil spill impact zones for post-spill Environmental Impact Assessment; assessed changes in phytoplankton and zooplankton community structure, species diversity indices, and biomass following hydrocarbon contamination.
- ❖ Collected, preserved, and taxonomically identified plankton samples, affected birds specimens from spill-affected marine environments; produced detailed technical species-level biodiversity reports for submission to CPCL and regulatory bodies.
- ❖ Contributed scientific data on plankton ecology to formal EIA documentation in compliance with Indian MoEF&CC environmental assessment standards; provided marine ecosystem restoration strategy recommendations.

Environmental Biologist — Marine EIA Biodiversity Baseline Survey

Chola MS Risk Services Pvt. Ltd., Chennai

2025 & 2026 | Karaikal Port, Southeast Coast of India & TNEB Baseline survey Tuticorin and Tirunelveli

- ❖ Conducted comprehensive biodiversity baseline surveys at Karaikal Port — systematic identification and enumeration of Birds, Reptiles, Mammals, Marine Invertebrates, and Insects for Marine EIA documentation.
- ❖ Conducted comprehensive flora and fauna biodiversity baseline surveys for Tamil Nadu Electricity Board (TNEB) transmission line project sites at Tuticorin and Tirunelveli for regulatory EIA submission.
- ❖ Developed conservation management plans and produced structured biodiversity baseline reports for coastal infrastructure regulatory compliance.

Project Associate — Marine Copepod Aquaculture (DBT-Funded)

Marine Planktonology & Aquaculture Laboratory, Bharathidasan University

April 2021 – May 2022 | Tiruchirappalli, India

Funded by Department of Biotechnology (DBT), India

- ❖ Designed and optimised high-density continuous culture systems for marine copepods (*Pseudodiaptomus annandalei, Dioithona rigida, Nitokra affinis*), simulating natural marine environmental conditions ex situ; contributed to resting egg (cyst) harvesting protocols.
- ❖ Peer-reviewed publication on continuous copepod production technology (*Biomass Conversion & Biorefinery*, Springer, 2023).

Ornithologist — Avian Ecology & Wildlife Hazard Management

Salim Ali Centre for Ornithology and Natural History (SACON), MoEF&CC

February 2021 – April 2021 | Coimbatore, India

Funded by Ministry of Environment, Forest and Climate Change, Government of India

- ❖ Conducted avian population monitoring and field assessments for bird hazard management, conservation planning, and habitat suitability analysis under a government-funded programme.

- ❖ Received specialist training in bird banding, mist netting, radio-tagging, camera traps, and behavioural ecology field protocols.

Environmental Educator & Field Coordinator

Pitchandikulam Forest Unit, Auroville Foundation

August 2020 – February 2021 | Pondicherry, India

- ❖ Coordinated wildlife surveys, biodiversity assessments, and community environmental education, trained field teams in ecological monitoring, GIS mapping, and data collection protocols.

M.Sc. Research Fellow — Urban Wildlife Ecology

Hajee Karutha Rowther Howdia College, Madurai Kamaraj University

August 2019 – February 2021 | Funded by Tamil Nadu State Council for Science and Technology (TNSCST)

- ❖ Investigated urban adaptation and population dynamics of House Sparrow (Passer domesticus) through systematic field surveys; produced peer-reviewed publication in Journal of Tropical Life Science (2021).

TAXONOMIC & ECOLOGICAL EXPERTISE

Marine Plankton Taxonomy

- ❖ Expert-level identification and enumeration of phytoplankton (diatoms, dinoflagellates, cyanobacteria, silicoflagellates) and zooplankton (copepods, chaetognaths, medusae, euphausiids, larval ichthyoplankton) using inverted, stereo zoom, and SEM microscopy with Sedgwick-Rafter and Bogorov tray counting methods.
- ❖ Taxonomic characterisation of copepod species including *Pseudodiaptomus annandalei, Acartia steueri*, and *Acartia* sp. through both wild sample identification and ex-situ culture work at Bharathidasan University.

Avian Ecology & Ornithology

- ❖ Qualified ornithologist (SACON, MoEF&CC-funded): species identification, population monitoring, avian hazard management, mist netting, bird banding, radio-tagging, and camera trap surveys.
- ❖ Conducted avian diversity surveys across coastal wetland, urban, and forest ecosystems, contributed to conservation planning and management reports, and presented research at Tamil Birders Meet.

Broad Taxonomy & EIA Biodiversity

- ❖ Full taxonomic coverage from Protozoa to Mammalia, applied in port-site biodiversity baseline surveys (Karaikal, 2024–25) identifying Birds, Reptiles, Mammals, Marine Invertebrates, Macro-algae, and Insects for ESIA regulatory documentation.

TECHNICAL SKILLS & INSTRUMENTATION

Artificial Intelligence & Machine Learning

- ❖ MATLAB Regression Learner: ANN, SVM, GPR, Ensemble Trees for PBM and ZBM prediction from oceanographic satellite data (SST, Chl-a); model performance metrics (R^2, RMSE, MAE).
- ❖ ANN model translation into SeaDAS band-math expressions for pixel-wise satellite biomass mapping.
- ❖ Python (NumPy, pandas, matplotlib) for data pre-processing and ML pipeline support.
- ❖ AI vision and Deep learning on Underwater Zooplankton Device.

Remote Sensing & GIS

- ❖ **SeaDAS**: OceanSAT-3 / OCM-3, MODIS-Aqua — SST, Chlorophyll-a, TSM ocean colour products; band math, pixel-wise algorithm validation.
- ❖ **QGIS & ArcGIS**: Spatial mapping, habitat distribution modelling, GPS-linked real-time field station logging.

Statistical & Visualization Tools

- ❖ SPSS, MINITAB, GraphPad Prism 9, SigmaPlot, Origin — ANOVA, multivariate regression, ecological trend analysis.

Laboratory & Analytical Instruments

- ❖ Inverted / Stereo Zoom / SEM Microscopy; GC-FID; UV-Vis Spectrophotometry; PCR and Gel Electrophoresis; sample preparation and preservation for plankton and DNA/RNA analysis.

Field & Oceanographic Equipment

- ❖ **Certified PADI Open Water Diver** – Facilitates submerged sampling and habitat assessment.
- ❖ Plankton Nets (WP-2, Bongo) — vertical hauls and oblique tows; CTD profilers; Niskin bottles; Secchi disc; GPS/GNSS field instruments.
- ❖ Mist nets and bird banding tools; radio-tagging; camera traps; light traps for insect population studies.

PEER-REVIEWED PUBLICATIONS

1. **Marjuk, M. S.**, Sarangi, R. K., Ayyasamy, G., Perumal, S., & Pachiappan, P. (2025). Modeling Phytoplankton and Zooplankton Biomass in the Southwestern Bay of Bengal: A Remote Sensing Approach with MODIS-Aqua and Oceansat-3 data. *Regional Studies in Marine Science*, 104425. https://doi.org/10.1016/j.rsma.2025.104425
2. **Marjuk, M. S.,** Sridhar, P., Gowthami, A., Santhanam, P., Sarangi, R. K., & Perumal, P. (2025). Systematic optimization and neural network modelling of *Acartia steueri* copepod reproduction for enhanced live feed solutions in sustainable aquaculture. *Aquaculture*, 743482. https://doi.org/10.1016/j.aquaculture.2025.743482
3. Gowthami, A., **Marjuk, M. S.,** Sarangi, R. K., Santhanam, P., & Perumal, P. (2026). Functional Trait Responses of Phytoplankton and Zooplankton to Environmental Variability in Nearshore Waters of the Bay of Bengal, Tamil Nadu Coast. *Continental Shelf Research*, 105704. https://doi.org/10.1016/j.csr.2026.105704
4. Sridhar, P., **Marjuk, M. S.,** Babu, T. D., Muralisankar, T., Santhanam, P., & Perumal, P. (2026). Influence of culture conditions and dietary regimes on survival, population dynamics and naupliar production of the copepod *Paramphiascella fulvofasciata* (Rosenfield & Coull, 1974). *Aquaculture International*, *34*(5), 144. https://doi.org/10.1007/s10499-026-02545-8
5. Santhanam, P., **Marjuk, M. S.,** Gunabal, S. et al. (2023). A novel technology towards the high-density and continuous production of the marine copepod, *Pseudodiaptomus annandalei* (Sewell, 1919). *Biomass Conversion and Biorefinery* (Springer). https://doi.org/10.1007/s13399-023-04348-w
6. Gowthami, A., **Marjuk, M. S.,** Santhanam, P., Thirumurugan, R., Muralisankar, T., & Perumal, P. (2025). Marine microalgae-mediated biodegradation of polystyrene microplastics: Insights from enzymatic and molecular docking studies. *Chemosphere*, 370, 144024. https://doi.org/10.1016/j.chemosphere.2024.144024
7. Gowthami, A., **Marjuk, M. S.**, Raju, P., Devi, K. N., Santhanam, P., Kumar, S. D., & Perumal, P. (2023). Biodegradation efficacy of selected marine microalgae against Low-Density Polyethylene (LDPE). *Marine Pollution Bulletin*, 190, 114889. https://doi.org/10.1016/j.marpolbul.2023.114889

8. Meeran, M., **Marjuk, S.,** Byrose, M., Arivoli, S., Tennyson, S., & Fathima, S. (2021). Population flux of the house sparrow *Passer domesticus* Linnaeus 1758 in Chinnamanur town, Theni district, Tamil Nadu. *Journal of Tropical Life Science*, 11(2). https://doi.org/10.11594/jtls.11.02.09

ENVIRONMENTAL IMPACT ASSESSMENT & CONSULTANCY

- ❖ EIA Biologist — Ennore Oil Spill Response (CPCL, Chennai Petroleum Corporation Limited, 2022–23 & TERI, Delhi): post-spill plankton biodiversity assessment, species-level biodiversity impact reporting, and marine restoration strategy recommendations.
- ❖ EIA Biologist — Karaikal Port Baseline Survey (Chola MS Risk Services Pvt. Ltd., Chennai, 2025): comprehensive biodiversity assessment for regulatory compliance covering Birds, Reptiles, Mammals, Marine Invertebrates, and Insects.
- ❖ EIA Biologist — TNEB Tuticorin & Tirunelveli project Baseline Survey (Chola MS Risk Services Pvt. Ltd., Chennai, Apr 2026): comprehensive biodiversity assessment for regulatory compliance covering complete fauna and flora.

AWARDS & RECOGNITION

1. **Dr. A.P.J. Abdul Kalam Young Research Award** — Terre Policy Centre (2022–23), presented 27 July 2023, for innovation and research excellence.
2. **M Krishnan Memorial Award for Nature Writing 2026 -** Madras Naturalists' Society, Chennai, Presented 04 May 2026,
3. **First Place** in **Oral Presentation** — National Conference on FutureBio 2025, Alagappa University (September 2024)
4. **Best Paper Presentation Award** — National Conference on Latest Trends in Life Sciences, HKR Howdia College (26 September 2019).
5. First Prize — ZooFest 2018 Zoology Quiz, Madurai American College.
6. **Wildlife Census Coordinator** — Tamil Nadu Forest Department wildlife census operations; led standardised population monitoring teams in coordination with forest officials and NGOs.
7. YouthForCop Climate Reality Leadership Certificate Programme (2025).
8. International Coastal Cleanup Day Volunteer — 2023, 2024, 2025 (Swachh Sagar Surakshit Sagar), Nagapattinam Beach, co-organised by Bharathidasan University & NCCR, MoES.
9. Class Representative for 5 consecutive years (B.Sc. & M.Sc.); served as Volunteer Coordinator at XLVI Indian Social Science Congress.

SELECTED CONFERENCES & PRESENTATIONS

1. FutureBio 2025 — Alagappa University: "Systematic Optimization and Neural Network Modelling of *Acartia steueri* Copepod Reproduction for Enhanced Live Feed Solutions in Sustainable Aquaculture" (September 2024).
2. NEAOsr'24 — Bharathidasan University: "Mass Production of Copepod *Acartia* sp. for Sustainable Aquaculture" (October 2024).
3. MARICON 2024 — Cochin University of Science and Technology (CUSAT), Kerala: International Conference on Frontiers in Marine Sciences (April 2024).
4. Blue Economy International Conference 2025 — Alagappa University Thondi (January 2025).
5. E-Content Development for Distance Education (E-CDDE-2023) — Bharathidasan University (April 2023).

6. National Conference on Environmental Sustainability, Health and Pollution Abatement (NCESHPA-2020) — Madurai (March 2020).
7. National Conference on Latest Trends in Life Sciences — HKR Howdia College, Theni (September 2019).

TRAINING PROGRAMMES & WORKSHOPS CO-ORGANISED

- ❖ Organized several seminars, scientific debates and quizzes during School, Bachelors, Masters, & PhD.
- ❖ Integrated Recirculating Marine Aquaponics (IRMA-2024) — Sponsored by TANCHE, Chennai; School of Marine Sciences, Bharathidasan University (March 2024).
- ❖ Mass Production of Marine Copepods and Cladocerans Culture (MPMCC-2022) — DBT-funded; Bharathidasan University (August 2023).
- ❖ Innovative Approaches for Sustaining Indian Fisheries and Aquaculture — Fisheries Technocrats Forum & Bharathidasan University (September 2023).
- ❖ Environmental Monitoring Using Advanced Analytical and Molecular Tools — DST-STUTI National Level Training Programme (November 2022).
- ❖ Marine Bioresources and Pollutant Biomarkers — DST-STUTI National Level Training Programme (October 2022).
- ❖ Mass Production of Marine Live Feeds Culture (MPMLF-2022); High-Density Continuous Production of Marine Copepods (HDCPMC-22); Sustainable Aquaculture Technologies (SAT-2021) — Bharathidasan University.

ORGANIZATIONAL AND SOCIAL SKILLS

- ❖ Organized several seminars, scientific debates and quizzes during School, Bachelors, Masters, & PhD.
- ❖ Played the master of ceremony (MC) for various departmental and cultural events.
- ❖ Experienced at working with students from various states of India and have experience in coordinating events for inter-college competitions and cultural events.
- ❖ Professional working experience with personnel from different ethnic backgrounds and multiple nationalities and taking care of welfare of all personnel involved.

PERSONAL DETAILS

❖ **Name**	:	SYED MARJUK M
❖ **Father's Name**	:	M. MOHAMMED KOYA
❖ **Mother's Name**	:	M. MALIKA BANU
❖ **Nationality**	:	INDIAN
❖ **Religion**	:	Muslim
❖ **Date of birth (Age)**	:	9th July 1998 (27 Years)
❖ **Sex**	:	Male
❖ **Marital Status**	:	Married
❖ **Blood Group (Rh)**	:	O (+)
❖ **Passport Number**	:	P9341457
❖ **Passport Validity**	:	10th Aprill 2027
❖ **Passport Issue Place**	:	Madurai, INDIA

REFERENCES

Dr. P. Santhanam MSc., PhD — ***PhD Supervisor*** Professor, Department of Marine Sciences, Bharathidasan University, Tiruchirappalli – 620 024, Tamil Nadu, India *Email:* santhanam@bdu.ac.in; santhanamcopepod@gmail.com	**Dr. N. Manoharan** MSc., MPhil., PhD Professor and Head, Department of Marine Science, Bharathidasan University, Tiruchirappalli – 620 024, Tamil Nadu, INDIA *Email:* biomano21@gmail.com
Mr. Srinivas R. Reddy, IFS Principal Chief Conservator of Forests & Chief Wildlife Warden, Government of Tamil Nadu Forest Department, Tamil Nadu, India. *Email:* reddysr66@rediffmail.com	**Dr. Venkata Mahalakshmi D.** MSc., PhD Scientist, Earth and Climate Sciences Area (ECSA), National Remote Sensing Centre (NRSC), Indian Space Research Organisation (ISRO), Hyderabad – 500 037, Telangana, India. *Email:* mahalakshmidv@nrsc.gov.in

DECLARATION

I hereby certify that all information provided in this Curriculum Vitae is accurate and truthful to the best of my knowledge and belief.

Place: Tiruchirappalli, India

Date: 10 August 2026

M. SYED MARJUK